Advanced Econometrics and Causal Inference

ECON 401 Midterm Examination • Time Allowed: 90 Minutes

Student Full Name: ______________________ **Student ID:** ____________
Section / Lab: ______________________ **Date:** August 28, 2026

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1. **Conceptual Econometric Foundations (15 Points)**

(a) (5 points) Suppose a linear regression model satisfies all Gauss-Markov assumptions except homoskedasticity (i.e., $\text{Var}(\varepsilon_i \mid \mathbf{X}_i) = \sigma_i^2$). What are the statistical properties of the OLS estimator $\hat{\beta}$?

A. Biased and inconsistent.

B. Unbiased and consistent, but no longer Best Linear Unbiased Estimator (BLUE).

C. Biased, but asymptotically efficient.

D. Unbiased and still asymptotically BLUE.

Solution: **Correct Choice: B.** Under heteroskedasticity, $\mathbb{E}[\hat{\beta} \mid \mathbf{X}] = \beta$ remains valid since $\mathbb{E}[\varepsilon \mid \mathbf{X}] = 0$. However, OLS is inefficient relative to Generalized Least Squares (GLS).

(b) (5 points) In a Two-Stage Least Squares (2SLS) model with instrument vector $\mathbf{Z}$, what condition defines instrument relevance?

A. $\mathbb{E}[\mathbf{Z}'\varepsilon] = \mathbf{0}$.

B. $\text{Rank}(\mathbb{E}[\mathbf{Z}'\mathbf{X}]) = K$, where K is the number of endogenous regressors.

C. $\mathbf{Z}'\mathbf{Z} = \mathbf{I}$.

D. The R^2 of the second stage must exceed 0.90.

Solution: **Correct Choice: B.** Instrument relevance requires non-zero correlation between instruments $\mathbf{Z}$ and endogenous regressors $\mathbf{X}$ (full column rank in the first-stage projection).

(c) (5 points) Under the parallel trends assumption in a Difference-in-Differences setup, what trajectory does the counterfactual estimate represent?

A. The control group if it had received treatment.

B. The treated group trajectory if it had not received treatment.

C. The overall sample population mean.

D. The instrumental variable projection.

Solution: **Correct Choice: B.** The counterfactual $\mathbb{E}[Y_{i2}(0) \mid D_i = 1]$ estimates the unobserved post-reform potential outcome of the treated group in the absence of treatment.

2. **Derivation of the Ordinary Least Squares (OLS) Estimator (20 Points)** Let $\mathbf{y} = \mathbf{X}\beta + \varepsilon$, where $\mathbf{y} \in \mathbb{R}^N$ and $\mathbf{X} \in \mathbb{R}^{N \times K}$ has full column rank.

   (a) (10 points) Write the Residual Sum of Squares (RSS) in matrix notation and compute the first-order condition with respect to β.

   Solution: $\text{RSS}(\beta) = (\mathbf{y} - \mathbf{X}\beta)^\top(\mathbf{y} - \mathbf{X}\beta) = \mathbf{y}^\top\mathbf{y} - 2\beta^\top\mathbf{X}^\top\mathbf{y} + \beta^\top\mathbf{X}^\top\mathbf{X}\beta$.
   Taking the gradient with respect to β and setting to zero:

   $$\frac{\partial \text{RSS}}{\partial \beta} = -2\mathbf{X}^\top\mathbf{y} + 2\mathbf{X}^\top\mathbf{X}\beta = \mathbf{0} \implies \mathbf{X}^\top\mathbf{X}\hat{\beta} = \mathbf{X}^\top\mathbf{y}$$

   (b) (10 points) Solve for $\hat{\beta}$ and derive its conditional variance $\text{Var}(\hat{\beta} \mid \mathbf{X})$ under homoskedasticity $\text{Var}(\varepsilon \mid \mathbf{X}) = \sigma^2\mathbf{I}_N$.

   Solution: Since $\mathbf{X}$ has full column rank, $(\mathbf{X}^\top\mathbf{X})^{-1}$ exists:

   $$\hat{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$$

   Substituting $\mathbf{y} = \mathbf{X}\beta + \varepsilon$:

   $$\hat{\beta} = \beta + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\varepsilon$$

   $$\text{Var}(\hat{\beta} \mid \mathbf{X}) = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\text{Var}(\varepsilon \mid \mathbf{X})\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1} \quad \blacksquare$$